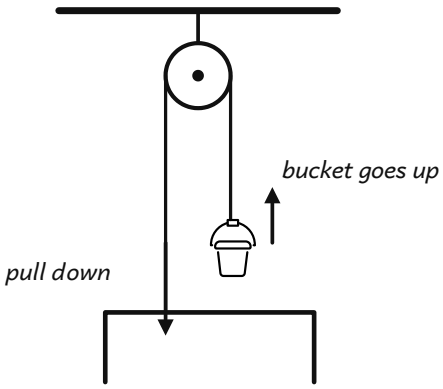
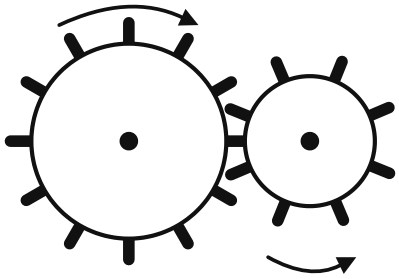
PHYSICAL AND CHEMICAL SCIENCE

Simple Machines

Levers, ramps, pulleys and gears, and how simple machines make work easier



pulley



gears

In this pack

- **Learn it:** Simple machines make work easier ★☆☆
- **Fact check:** Name the machine ★☆☆
- **Think it through:** How do machines help? ★★☆
- **Investigate:** Does a ramp make it easier? ★★☆
- **Real-world case study:** From dheki to rice mill ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C5 · SIMPLE MACHINES

Simple machines make work easier

★☆☆ Grade 3

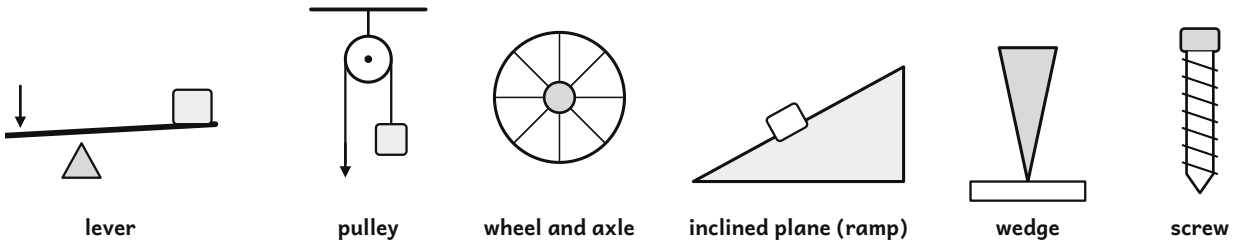
Name: **Class:** **Date:**

Learning goals

- Explain what a simple machine is.
- Find levers in everyday life and explain how they make work easier.
- Explain how ramps, pulleys and gears help us.

Try to lift the lid off a tin of powdered milk with your fingers. Hard! Now push the end of a spoon under the lid and press down. Easy! The spoon is a **simple machine**.

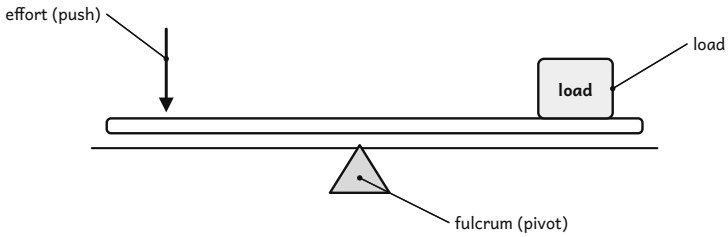
A **simple machine** is a tool with few or no moving parts that makes work **easier**. It may let us use **less force**, **change the direction** of a force, or move something **further or faster**.



Six simple machines.

Levers

A **lever** is a bar that turns on a point called the **fulcrum** (pivot). You push on one part (the **effort**) to move a **load**. The **further** your effort is from the fulcrum, the **less force** you need.

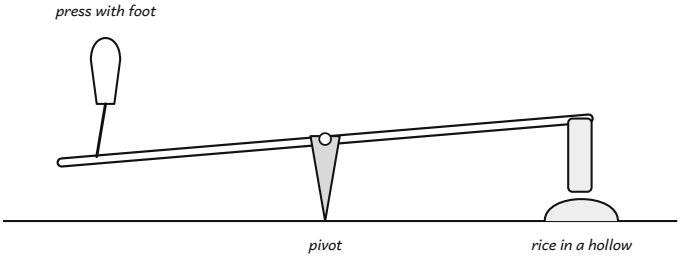


A see-saw lever.

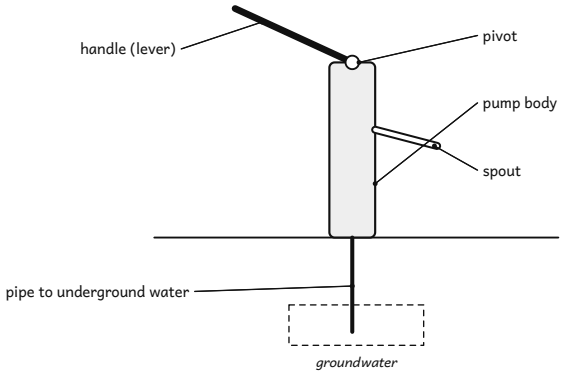
scissors	see-saw in a park	spoon opening a lid	wheelbarrow	hammer pulling a nail

Bangladesh connection

A **dheki** is a long wooden lever used in villages to husk rice, and a **tubewell** handle is a lever that lifts water.



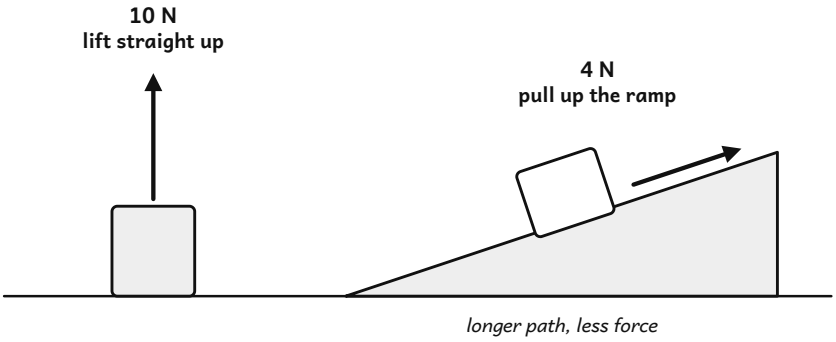
A dheki.



A tubewell.

Ramps (inclined planes)

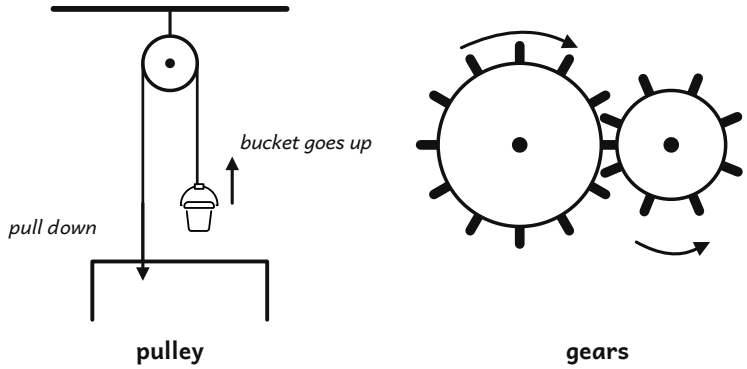
A **ramp** is a slope. It is easier to push a heavy load up a ramp than to lift it straight up: you move it **further**, but with **less force**. Ramps help wheelchair users, people loading trucks, and cars on hill roads that zig-zag up the slope.



A ramp means less force.

Pulleys and gears

A **pulley** is a wheel with a rope over it. Pulling **down** on the rope lifts the load **up**. Pulleys lift buckets from wells, flags up poles and heavy loads on building sites. **Gears** are wheels with teeth that turn each other. Bicycles, clocks, egg beaters and rice mills use gears.



A pulley and two gears.

Key words

simple machine	a tool that makes work easier
lever	a bar that turns on a fulcrum
fulcrum	the point a lever turns on
effort	the push or pull you give
load	the thing being moved
pulley	a wheel with a rope
gear	a wheel with teeth

FACT CHECK · C5 · SIMPLE MACHINES

Name the machine

★☆☆
Grade 3

Name: _____ **Class:** _____ **Date:** _____

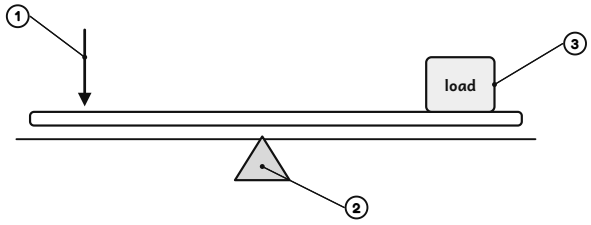
Learning goals

- Label a lever.
- Name simple machines.

1 Label the lever. [3]

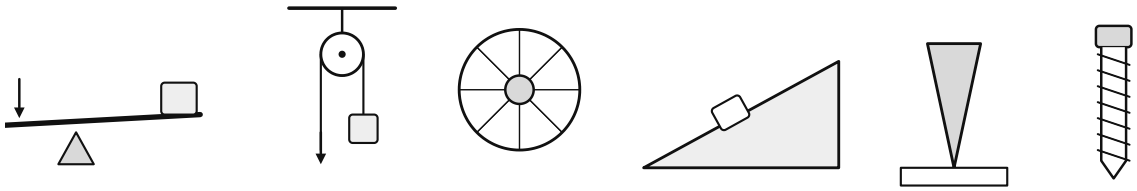
LABELS

fulcrum (pivot) load effort (push)



1 _____ 2 _____ 3 _____

2 Name each simple machine. [6]



A B C D E F

A	B	C	D	E	F

3 Draw a line to match. [4]

Draw a line to match each pair, or write the letter.

scissors ●	● **A** lever
wheelchair slope ●	● **B** pulley
flag pole rope ●	● **C** ramp (inclined plane)
bicycle chain wheels ●	● **D** gears

THINK IT THROUGH · C5 · SIMPLE MACHINES

How do machines help?

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Explain how machines make work easier.

1 **Why is it easier to open a tin lid with a long spoon than with a coin?** [2]

2 **Why do roads up steep hills zig-zag?** [2]

3 **How does a pulley help someone raise a flag?** [2]

4 **Draw a lever you use at home. Mark the fulcrum, effort and load.** [3]

My lever

INVESTIGATE · C5 · SIMPLE MACHINES

Does a ramp make it easier?

★★☆ Grade 3

Name: Class: Date:

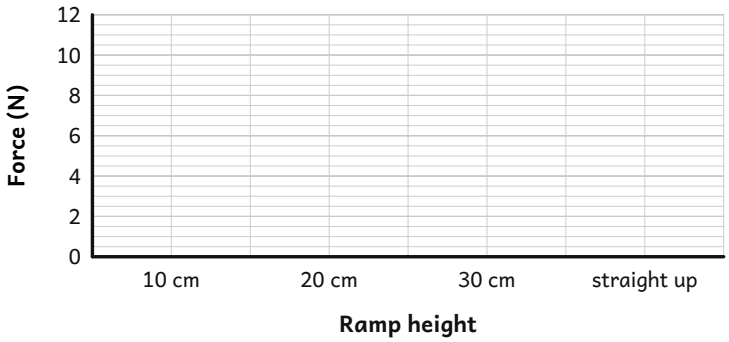
Learning goals

- Measure force in newtons.
- Draw a bar chart and conclude.

Class 3C used a forcemeter to pull a toy truck up ramps of different heights. Lifting it straight up needed 10 N.

1 **Draw a bar chart.** [3]

Ramp height	10 cm	20 cm	30 cm	straight up
Force (N)	3	5	7	10



2 **Which ramp needed the least force?** [1]

3 **Complete: the lower (gentler) the ramp, the ___ force is needed.** [1]

4 **What did they keep the same?** [1]

REAL-WORLD CASE STUDY · C5 · SIMPLE MACHINES

From dheki to rice mill

★★★ Grade 3

Name: **Class:** **Date:**

Learning goals

- See how machines change daily life.

Science | History | Work

For hundreds of years, village women in Bangladesh husked rice with a **dheki**. One person stepped on the end of a long wooden beam. The beam turned on a pivot and the other end lifted, then thumped down onto the rice to knock off the husks. It was slow, hard work.

Today most rice is husked in **rice mills**. A motor turns **gears** and **belts** that work the machines, and one mill can husk tonnes of rice in a day. Some families still use a dheki to make special rice flour for winter pitha.

Based on rice husking in Bangladesh.

1 **What kind of simple machine is a dheki?** [1]

2 **How did the dheki make work easier?** [2]

3 **Write one good thing and one problem about rice mills.** [2]

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use machine words.

1 Find the words.

N	G	M	L	E	V	E	R	H
L	E	A	C	H	L	Y	G	H
P	A	E	F	F	O	R	T	T
D	R	I	R	A	K	P	K	V
P	U	L	L	E	Y	H	N	F
K	W	W	H	E	E	L	R	G
L	O	A	D	W	Y	R	A	F
D	Y	G	P	S	V	L	U	S
H	F	F	B	V	R	A	M	P

effort gear lever load pulley ramp wheel

2 The point a lever turns on is the... [1]

◯ load ◯ fulcrum

◯ gear

3 A ramp lets you use... [1]

◯ more force ◯ less force

◯ no force

4 Challenge. Design a machine to lift a bucket of sand to a first-floor window. Label the simple machines you used. [3]

My machine

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain what a simple machine is	○	○	○
find the fulcrum, effort and load	○	○	○
explain how a ramp helps	○	○	○
explain how pulleys and gears help	○	○	○
measure force on a ramp	○	○	○